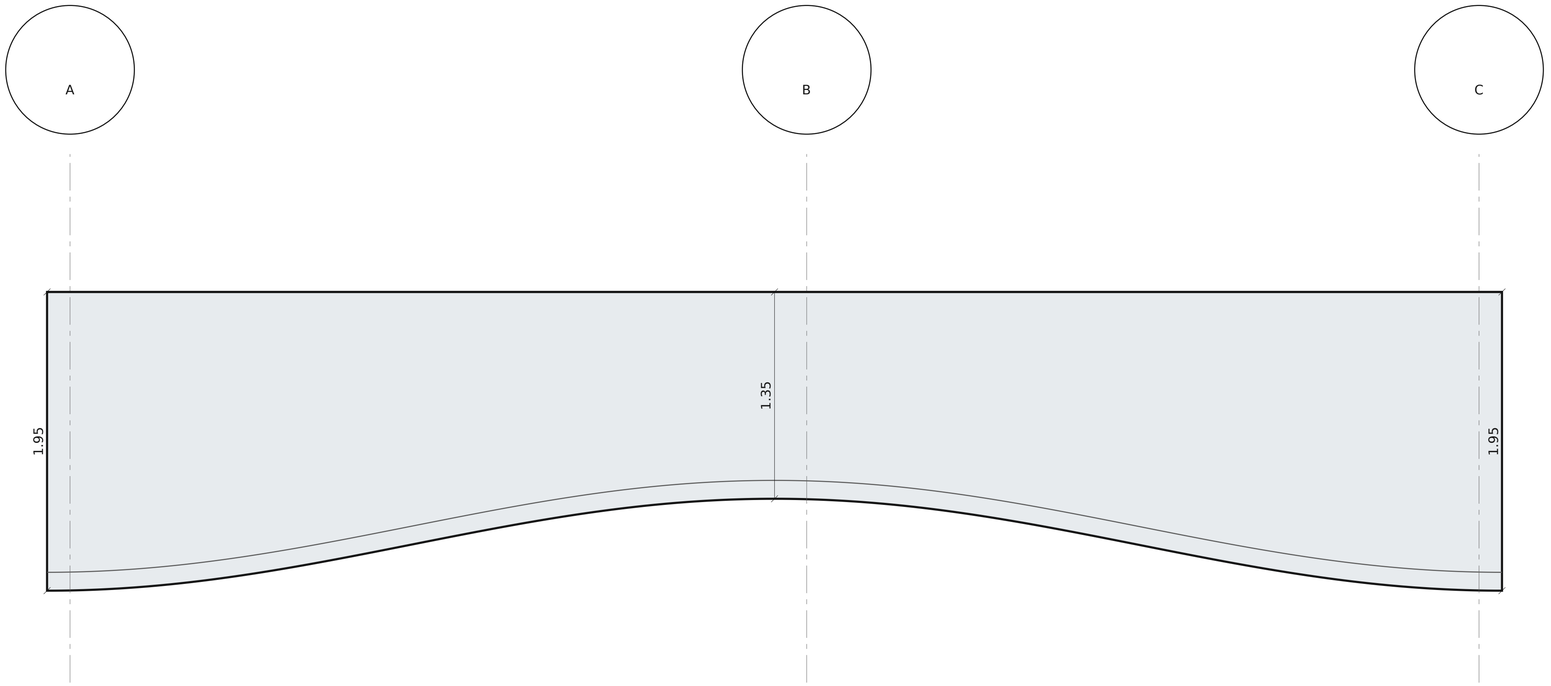

SLAB EDGE — THICKNESS, ROOT DEPTH AND REINFORCEMENT BY THE ENGINEER

CANTILEVER BALCONY STRUCTURE

LEVEL +13.20 and the three floors below · 1:20 @ A1 · METRES

KEY

- External / party wall cut by section, 300 mm — brick, colour unrestricted
- Internal partition, 100 mm — red brick
- Terrace parapet, 250 mm × 1100 mm high
- Single leaf door, swing shown to opening side
- Glazed opening — arched head where noted
- Element above the cut plane (pergola, slab over)
- Void / open to sky
- +0.00 Finished floor level datum

THE FRAME

PROJECTION, DEEPEST	1.95 m
PROJECTION, AT THE WAIST	1.35 m
SPAN OF THE CURVE	9.50 m
AREA A FLOOR	15.7 m²
LEVELS	4
TOTAL	62.7 m²
BALUSTRADE	glass, 120 mm inboard
ROOT AND REINFORCEMENT	by the engineer

GENERAL NOTES

1. Layout only. No member size, no slab or footing depth, no reinforcement and no load appears on this sheet or anywhere in the STR series. Those are the structural engineer's, and a column schedule with sizes invented by an architect is worse than none, because somebody builds it.

2. The grid, the column coordinates, the spans and the slab edges are measured from the same geometry the architecture is drawn from and are exact.

3. A 1.95 m cantilever on a curve, repeated on four levels, is the single most engineering-sensitive element in this building. It is drawn at 1:20 for that reason and for no other.

4. The curve, the span, the projection at every point and the balustrade line are exact. The slab thickness, the root depth into the floor behind, the top steel and the deflection and vibration checks are entirely the engineer's.

5. Grid line 1 runs along the face the cantilever springs from. The balcony projects south of it and is unsupported at its far edge on all four levels.

FIVE-STOREY VACATION VILLA — EZBIT ABU RUKBAH, QUESNA, MENOUFIA, EGYPT

DRAWING

CANTILEVER BALCONY STRUCTURE

DRAWING No	REVISION	SCALE	DATE
STR-500	**P30**	**1:20 @ A1**	**2026-10-12**

Layout for the engineer's design — no member sizes, no reinforcement, no loads. Not for construction.
Do not scale; work to figured dimensions only.

REVISIONS

P27 2026-10-11 the stair, which the interior cutaways did not show at all
P28 2026-10-11 ARC-400 centred: the last sheet still on the old wide origin
P29 2026-10-11 STR-200: the column schedule it was named for; heights corrected
P30 2026-10-12 site corrected to Ezbit Abu Rukbah; Code 305 named as the fire code